

Corollary 4.2. *Assume X has a shrinking F. D. D., $\mathcal{A} \in L_s^1(\Omega, \mu, \mathcal{B}(X))$ is a commuting family of compact operators. Then, we have the following inequality*

$$r\left(\int_{\Omega} \mathcal{A} d\mu\right) \leq \int_{\Omega} r(\mathcal{A}_t) d\mu(t).$$

Moreover, if $\mathcal{A}_t$ are Volterra operators, so is $\int_{\Omega} \mathcal{A} d\mu$.

Proof. By Proposition 3.2 we get that $\mathcal{A} : \Omega \rightarrow \mathcal{B}(X)$ is strongly μ -measurable function. Furthermore, as X^* is separable (because X has a shrinking F. D. D.) it cannot contain an isomorphic copy of ℓ^∞ and thus cannot contain an isomorphic copy of $\mathfrak{c}_0$, due to [12, Corollary 6, Page 23]. Final conclusion now follows from Theorem 4.1. If $\mathcal{A}_t$ is a Volterra operator for all $t \in \Omega$, then due to proved inequality we have that the operator $\int_{\Omega} \mathcal{A} d\mu$ is Volterra too. $\square$

Concluding remarks: It is not known to the authors whether the strong μ -measurability of the family $\mathcal{A}$ in Theorem 4.1 can be replaced by a weaker assumption. We also note that Lemma 4.4 provides alternative conditions for μ -measurability of function $\Omega \ni t \mapsto r(\mathcal{A}_t) \in \mathbb{R}$.

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REFERENCES

- [1] C. D. Aliprantis, K. C. Border. *Infinite Dimensional Analysis*, Springer, Third edition, (2007).
- [2] Arsenović, M., Bogachev, V.I. Krstić, M. *Spaces of Measure-Valued Mappings Connected with Disintegrations*, Sib Math J 66, 248–261 (2025).
<https://doi.org/10.1134/S0037446625020028>
- [3] Miloš Arsenović, V.I. Bogachev, Mihailo Krstić, *Integration of functions with values in spaces of measures*, Proc. Steklov Inst. Math., 2025, Volume 331, pages 1–27, (2025).
<https://doi.org/10.1134/S0081543825601534>
- [4] Arsenović, M., Krstić, M. *A Normed Space of Weakly Integrable Operator-Valued Functions and Convergence Theorems*, Bull. Iran. Math. Soc. 51, 30 (2025).
<https://doi.org/10.1007/s41980-024-00965-x>
- [5] Bessaga, C. and Pelczyński, A., *On bases and unconditional convergence of series in Banach spaces*, Studia Math., 17, (1958) 151–164.
- [6] Blasco Oscar, and Jan van Neerven. *Spaces of operator-valued functions measurable with respect to the strong operator topology*. Vector measures, integration and related topics. Basel: Birkhäuser Basel, 2010. 65–78.
https://doi.org/10.1007/978-3-0346-0211-2_6
- [7] Blasco Oscar, and Ismael Garcia Bayona. *Remarks on Measurability of Operator-Valued Functions*. Mediterranean Journal of Mathematics, Volume 13, pages 5147–5162, (2016).
<https://doi.org/10.1007/s00009-016-0798-1>
- [8] S. Bochner, *Integration von Funktionen, deren Werte die Elemente eines Vektorraumes sind*, Fundamenta Mathematicae, vol. 20, pp. 262–276, 1933.
<https://eudml.org/doc/212635>